GEOSPATIAL PROXIMITY AUDIO ECOSYSTEM

✓ 100% MODULE COMPLETED

Module 01: Proximate & Live Spatial Audio Engine

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Repository:	amandubey0907/My-first-site-	**Review Status:**	Ready for Faculty / Guide Evaluation

1. EXECUTIVE SUMMARY & CORE OBJECTIVES

This module implements a real-time, browser-native **Spatial Proximity Sound Engine**. It simulates real-world acoustics where sound volume dynamically attenuates with distance according to acoustic decay laws, stereo pans between ears based on relative user angle, and automatically cuts off at boundary distances to save bandwidth and compute resources.

2. DETAILED PROGRESS BREAKDOWN & VERIFICATION

Task ID	Component / Deliverable	File / Architecture	Status	Verification & Proof
T-01	**Euclidean & Haversine Distance**	`proximityEngine.js`	COMPLETED	Earth radius formula (R = 6,371 km) & 2D coordinates.
T-02	**3 Acoustic Decay Curves**	`proximityEngine.js`	COMPLETED	Inverse (1/r), Linear & Exponential with rolloff factor.
T-03	**Dynamic Stereo Panning**	`proximityEngine.js`	COMPLETED	Normalized bearing factor (-1.0 Left to +1.0 Right).
T-04	**Web Audio API DSP Graph**	`audioEngine.js`	COMPLETED	Gain & Panner nodes chained to Master and Destination.
T-05	**Harmonic Synthesizer Beacon**	`audioEngine.js`	COMPLETED	Dual oscillators (220Hz, 330Hz, 440Hz) for instant testing.
T-06	**Live Microphone Stream Engine**	`audioEngine.js`	COMPLETED	WebRTC getUserMedia with echo & noise suppression.
T-07	**Live FFT Spectrum Visualizer**	`audioEngine.js`	COMPLETED	64-bin FFT canvas animation with linear gradient styling.
T-08	**2D Radar Viewport & HUD**	`app.js` / `index.html`	COMPLETED	60 FPS canvas loop with live Distance, Gain % & Pan display.
T-09	**Keyboard WASD & Drag Controls**	`app.js`	COMPLETED	Smooth multi-key WASD / Arrow keys with Shift sprint boost.
T-10	**Interactive Presentation Presets**	`app.js` / `index.html`	COMPLETED	Core (100%), Mid Decay, Cutoff (0%), Left/Right Ear gliding.

3. WEB AUDIO API SIGNAL FLOW GRAPH



4. MATHEMATICAL ACOUSTIC FORMULATIONS

1. Haversine Distance (GPS Spherical Geometry):

$$a = \sin^2(\Delta lat / 2) + \cos(lat1) \cdot \cos(lat2) \cdot \sin^2(\Delta lon / 2)$$
$$d = 2 \cdot R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{(1 - a)}) \quad [\text{where } R = 6{,}371{,}000 \text{ meters}]$$

2. Inverse Distance Attenuation Curve:

$$Gain(d) = d_{ref} / (d_{ref} + rolloff \cdot (d - d_{ref}))$$

Boundary Conditions: If $d \le d_{ref} \rightarrow Gain = 1.0$ (Core 100%) | If $d \ge d_{max} \rightarrow Gain = 0.0$ (Silent Cutoff)

3. Normalized Stereo Panning Bearing:

$$Pan_factor = clamp((beacon_x - listener_x) / \max(0.1, distance), -1.0, +1.0)$$

5. GIT COMMIT AUDIT TRAIL (BRANCH: AMAN--FEATURE)

Commit Hash	Commit Message	Branch	Remote Sync
`fceaaa6`	chore: update author name to Aman Dwivedi across UI, radar canvas and project documentation	`aman--feature`	SYNCED
`3d5280b`	docs: add formal project progress sheet and guide evaluation rubric	`aman--feature`	SYNCED
`9fcc14f`	feat(presentation): add interactive demo presets and smooth glide animation for live presentations	`aman--feature`	SYNCED
`492c697`	feat(navigation): add keyboard WASD and arrow controls for radar navigation	`aman--feature`	SYNCED
`4308950`	feat(audio): complete geospatial proximity audio and live spatial sound engine	`aman--feature`	SYNCED

6. Guide / Faculty Milestone Evaluation Form

Criterion	Max Marks	Marks Awarded	Guide Remarks
Mathematical Distance & Attenuation Logic	20		
Web Audio API Implementation & Stereo Panning	20		
Interactive Radar Canvas & Visualizer Feedback	20		
Live Microphone Stream & Real-World Utility	20		
Code Quality, Documentation & Git Versioning	20		
Total Score	**100**		

Student Signature (Aman Dwivedi)	Project Guide Signature (Faculty Evaluator)	Date of Review (DD / MM / YYYY)